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# Single-Source Shortest Paths: From Dijkstra to DMMSY

**A CMSC 142 portfolio study comparing five SSSP algorithms in pure-Python stdlib, with DMMSY (2025) as the centerpiece.**

**Zildjian E. California**[1] **Rey Marvin C. Rizal**[1]

1 Department of Mathematics, Physics, and Computer Science,
College of Science and Mathematics, University of the Philippines - Mindanao

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# Outline

# Problem and Motivation

## Problem

Given a weighted graph $G = (V, E)$ with edge weights $w\colon E \to \mathbb{R}$ and a source vertex $s \in V$, compute:

$$d(v) = \min_{\pi : s \to v} \sum_{e \in \pi} w(e) \qquad \text{for every } v \in V.$$

- Edge orientation: directed for Dijkstra, Bellman–Ford, A*, DMMSY; undirected for Thorup.
- Weight class: non-negative reals for Dijkstra / A* / DMMSY; signed reals for Bellman–Ford; non-negative integers for Thorup.
- Output: a distance map $d\colon V \to \mathbb{R}$ (and, optionally, a predecessor map for path reconstruction).

- **Dijkstra (1959)** [1] – the foundational non-negative-weight algorithm; binary-heap relaxation.
- **Bellman–Ford (1958)** [2], drawing on Ford (1956) [3] – handles negative weights; detects negative cycles.
- **A* (1968)** [4] – heuristic-guided search; degenerates to Dijkstra at $h \equiv 0$.
- **Thorup (1999)** [5] – linear-time undirected SSSP for non-negative integer weights in the word-RAM model.
- **DMMSY (2025)** [6] – deterministic $O(m \log^{2/3} n)$ directed SSSP; the centerpiece of this study.

We re-implement all five in pure-stdlib CPython 3.11+ and benchmark each against Dijkstra on a documented suite.

# Baseline Algorithms

# Dijkstra (1959) – Heap Relaxation

## Problem Class

Non-negative weighted directed (or undirected) graphs; single source $s$.

## Design Idea

Maintain a frontier set $B$ in a min-heap keyed by tentative distance; repeatedly extract the closest unsettled vertex, settle it, and relax its outgoing edges.

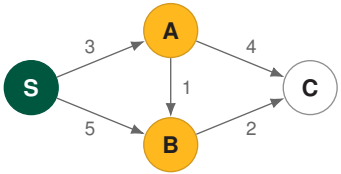
## Complexity

Time $O((n+m)\log n)$; space $O(n)$.

## Implementation

`src/sssp/dijkstra.py`;
`heapq`-backed;
7 tests, 100% line coverage.
Source paper: [1].

# Dijkstra – A Worked Heap Relaxation




**Step 1: pop S (d=0)**

| | |
|---|---|
| settled | {S:0} |
| heap | [(3,A),(5,B)] |
| relaxed | S→A, S→B |

**Step 2: pop A (d=3)**

| | |
|---|---|
| settled | {S:0, A:3} |
| heap | [(4,B),(5,$B^*$),(7,C)] |
| relaxed | A→B (5→4), A→C |

**Step 3: pop B (d=4)**

| | |
|---|---|
| settled | {S:0, A:3, B:4} |
| heap | [(5,$B^*$),(6,C),(7,C)] |
| relaxed | B→C (7→6) |

**Step 4: pop C (d=6) – done**

| | |
|---|---|
| settled | {S:0,A:3,B:4,C:6} |
| heap | stale entries skipped |
| result | all distances final |

■ settled ■ frontier $B^*$ = stale entry, ignored on pop

# Bellman–Ford (1958, Ford 1956) – Edge Relaxation

## Problem Class

Directed graphs with possibly negative edge weights; detect negative-weight cycles reachable from $s$.

## Design Idea

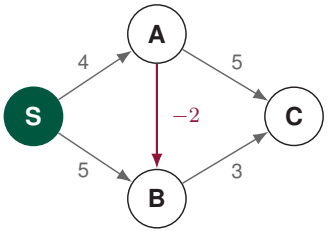
Repeat full-graph edge relaxation $n - 1$ times; one extra pass diagnoses any reachable negative cycle.

## Complexity

Time $O(nm)$; space $O(n)$.

## Implementation

`src/sssp/bellman_ford.py`; reports `NegativeCycleReport`; 100% line coverage.
Source papers: [2], [3].




**Edge scan order**

1. (S,A,+4)
2. (S,B,+5)
3. (A,B,−2) ← key relaxation
4. (B,C,+3)
5. (A,C,+5)

**Distance after each round**

| vertex | init | R1 | R2 |
|---|---|---|---|
| S | 0 | 0 | 0 |
| A | ∞ | 4 | 4 |
| B | ∞ | $5 \to 2$ | 2 |
| C | ∞ | 5 | 5 |

- Each round sweeps every edge once; relaxing $(A, B, -2)$ *after* $(S, B, 5)$ lowers $d[B]$ from 5 to 2.
- R2 produces no update ⇒ early-exit; the final cycle-check pass also relaxes nothing ⇒ `has_cycle = False`.

# A* (1968) – Heuristic Guidance

## Problem Class

Goal-directed shortest path with a domain-specific heuristic $h(v)$ lower-bounding $d(v, \text{goal})$.

## Design Idea

Order the frontier by $f(v) = g(v) + h(v)$ instead of $g(v)$ alone; admissible $h$ preserves optimality.

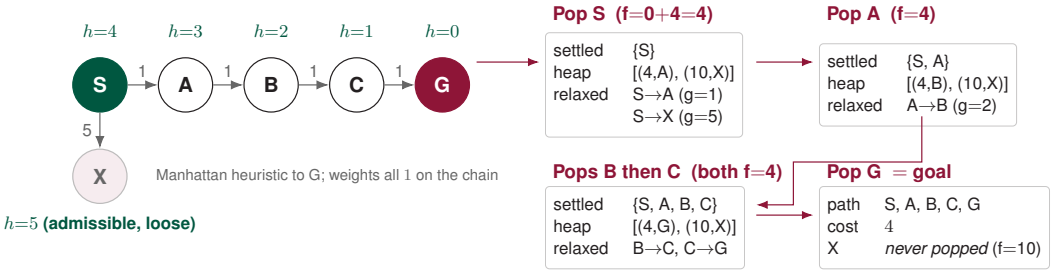
## Complexity

Time $O((n + m) \log n)$ worst case; expands fewer nodes than Dijkstra when $h$ is informative.

## Implementation

`src/sssp/astar.py`, `src/sssp/heuristics.py`; ships `manhattan`, `zero`; $A^*(h \equiv 0) \equiv$ Dijkstra tested. Source paper: [4].

# A* – A Worked Heuristic-Guided Search




- Every chain vertex has $f = g + h = 4$, so A* pops them in order and never expands the dead-end branch X (its $f = 5 + 5 = 10$).
- Set $h \equiv 0$ and A* degenerates to Dijkstra – the equivalence is asserted as a correctness test in `tests/test_astar.py`.

# Thorup (1999) – Linear-Time Undirected SSSP

## Problem Class

Undirected graphs with non-negative integer edge weights; word-RAM model.

## Design Idea

Hierarchical bucket structure exploiting integer weight bits and constant-time word operations to avoid the sorting bottleneck of comparison-based SSSP.

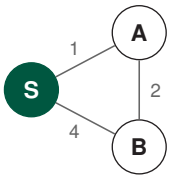
## Complexity

Time $O(n + m)$ in the word-RAM model.

## Implementation

`src/sssp/thorup99.py`; 11 tests, 99% line coverage. *Python* $\neq$ *word-RAM:* caveat documented in the project notes. Source paper: [5].

# Thorup – A Worked MSDB Bucket Walk




undirected, integer weights

**After popping S**

**last_min = 0**

**bucket 1** A : $d$=1

**bucket 3** B : $d$=4

*msdb*$(1,0)$=1; *msdb*$(4,0)$=3

**Advance, settle A, relax**

**last_min = 1**

**bucket 0** A settled

**bucket 2** B : $4 \to 3$

**bucket 3** B* stale

**Advance, settle B**

**last_min = 3**

**bucket 0** B settled

**result** $d=\{S{:}0, A{:}1, B{:}3\}$

*stale B* skipped on pop*

- Buckets are keyed by $\mathrm{msdb}(d(v),\ last\_min)$ – the most-significant differing bit between a tentative label and the current minimum.
- Advancing `last_min` re-bins live entries; lazy-stale entries (B*) are skipped on pop. Word-RAM gives $O(m+n)$; CPython does not.

# DMMSY (Centerpiece)

## Setting

Directed SSSP with non-negative real weights; labels use only comparison and addition.

## Barrier

Dijkstra-style methods still pay for global priority ordering, giving the sparse-graph comparison baseline $O(m + n \log n)$.

## DMMSY Claim

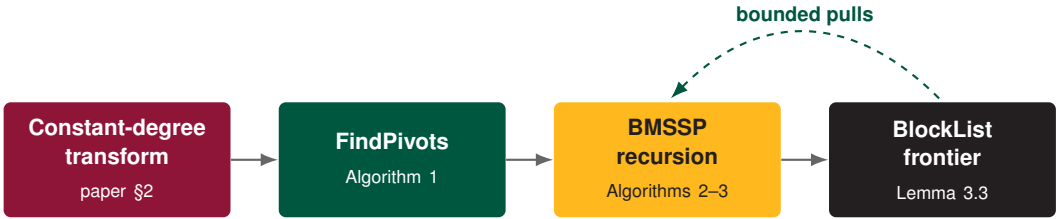
Duan–Mao–Mao–Shu–Yin break the barrier deterministically:

$$O\left(m \log^{2/3} n\right).$$

## Portfolio Role

Centerpiece algorithm.
Source paper: [6].

Implementation:
`src/sssp/dmmsy/`.




- **Transform:** reduce high-degree vertices so work charges uniformly per arc.
- **FindPivots:** shrink the frontier to useful recursion roots.
- **BMSSP + BlockList:** solve bounded subproblems without globally sorting every label.

## Contract

BMSSP$(G, \ell, B, S, d, k, t)$ solves labels below boundary $B$ from source set $S$, then returns a smaller boundary $B'$ and settled set $U$.

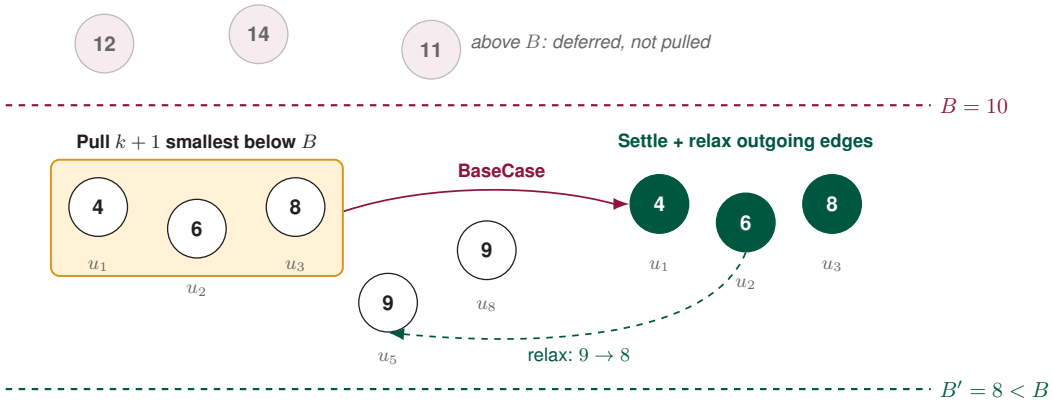
## Level 0

BASECASE: bounded Dijkstra from one source, stopping after $k + 1$ discoveries or boundary exhaustion.

## Level > 0

1. FINDPIVOTS shrinks $S$ to pivots $P$.
2. BlockList pulls bounded batches below $B$.
3. Recursive calls settle batches and relax outgoing edges.

**Key idea:** local bounded batches replace one global sorted frontier.




- Pull only the $k+1$ smallest tentative labels below $B$ – no global sort.
- Settle, relax, return a tighter boundary $B' < B$; deferred labels stay above $B$.

## Paper Parameterization

$$k = \left\lfloor (\log_2 n)^{1/3} \right\rfloor$$

$$t = \left\lfloor (\log_2 n)^{2/3} \right\rfloor$$

$$\ell_{\max} = \left\lceil \frac{\log_2 n}{t} \right\rceil$$

Balances pivot work, recursion depth, and block-list operations.

## Asymptotic Result

Deterministic directed SSSP:

$$O\left(m \log^{2/3} n\right) \text{ time}$$

Space stays $O(n + m)$.

## Python Implementation Boundary

CPython recursion overhead is real. Default large numeric benchmark runs use the documented shortcut; `Weight` inputs and explicit parameters exercise recursive BMSSP.

# Experimental Evaluation

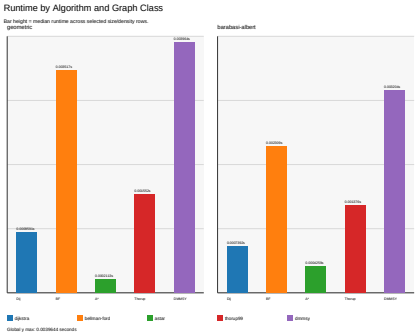
# Benchmark Protocol

## Dataset

Seed 2026; 12 benchmark cases; geometric and Barabasi–Albert families at $n \in \{64, 256, 1024\}$ with sparse/dense profiles.

## Timing

60 CSV rows; setup and file IO excluded; one recorded repeat per case-algorithm pair; Dijkstra is the normalization baseline.



## Median runtime ratio by size bucket

| Algorithm | Small $n = 64$ | Medium $n = 256$ | Large $n = 1024$ |
|---|---|---|---|
| Dijkstra | 1.00× | 1.00× | 1.00× |
| Bellman–Ford | 1.92× | 3.48× | 3.87× |
| A* | 0.80× | 0.29× | 0.50× |
| Thorup (1999) | 1.84× | 1.61× | 1.36× |
| DMMSY (2025) | 11.28× | 5.87× | 1.02× |

- A* benefits from an informative geometric heuristic; Bellman–Ford pays for repeated edge sweeps.
- Thorup stays above Dijkstra because CPython is not the word-RAM model.
- DMMSY passes the large-size runtime sanity check: max DMMSY/Dijkstra ratio is `1.093x` for $n \geq 1000$.

## Notable Artifacts

- Github repository: `https://github.com/zcalifornia-ph/sssp-modern`
- CSV: `bench/results/sssp-benchmark-seed-2026.csv`
- Figure: `bench/figures/runtime-by-algorithm.{pdf,png}`
- Tests: 147-test suite passed after benchmark plot generation; the runtime check records max DMMSY/Dijkstra ratio `1.093x` for $n \geq 1000$.

## Final Words

The portfolio centers the algorithmic result and implementation structure, not a CPython timing victory: correctness and comparison-addition fidelity are validated separately from the fast-path benchmark policy.

# Reflection and Author Contributions

## What The Comparison Teaches

No SSSP algorithm dominates every setting; the right choice depends on weight signs, direction, integer assumptions, heuristic quality, and runtime model.

## Honest DMMSY Framing

The centerpiece is the deterministic $O(m \log^{2/3} n)$ comparison-addition bound. CPython does not show a recursive BMSSP wall-clock win; large numeric runs use the documented shortcut to keep the runtime check passing.

## AI Assistance Disclosure

AI tools assisted with drafting, tests, supporting files, and prose. The source papers, validation evidence, and final judgments remain the human authors' responsibility.

# Author Contributions

## Zildjian E. California

- Repository foundation, root documentation, graph/weight/IO/generator base, Dijkstra, A*, DMMSY, benchmarks, report, and deck authorship.
- DMMSY centerpiece: transform, block list, pivots, BMSSP, driver, written rationale, and presentation opening/centerpiece.
- Current commit history: authored the foundation, Dijkstra/A*, DMMSY stack, benchmark files, report manuscript, and deck scaffold.

## Rey Marvin C. Rizal

- Bellman–Ford implementation with negative-cycle reporting; assigned contribution completed.
- Thorup 1999 runtime-model work and hierarchical-bucket SSSP contribution; assigned contribution completed.
- Presentation ownership: Bellman–Ford, Thorup, and experimental-results delivery; joint Q&A with California.

Contribution split follows the GitHub repository commit history and the closed issues and pull requests.

# Questions?

Zildjian E. California github.com/zcalifornia-ph
Rey Marvin C. Rizal github.com/marverickdev

https://github.com/zcalifornia-ph/sssp-modern

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The five source papers cited above are listed below; the Beamer class [7] and the UP Visual Identity Guidebook [8] are credited for the deck infrastructure.

[1] E. W. Dijkstra. “A Note on Two Problems in Connexion with Graphs”. In: *Numerische Mathematik* 1.1 (1959), pp. 269–271. DOI: 10.1007/BF01386390.

[2] Richard Bellman. “On a Routing Problem”. In: *Quarterly of Applied Mathematics* 16.1 (1958), pp. 87–90. DOI: 10.1090/qam/102435.

[3] L. R. Ford. “Network Flow Theory”. In: P-923 (1956). Also available at https://www.rand.org/pubs/papers/P923.html. URL: https://apps.dtic.mil/sti/tr/pdf/AD0422842.pdf.

[4] Peter E. Hart, Nils J. Nilsson, and Bertram Raphael. “A Formal Basis for the Heuristic Determination of Minimum Cost Paths”. In: *IEEE Transactions on Systems Science and Cybernetics* 4.2 (1968), pp. 100–107. DOI: 10.1109/TSSC.1968.300136.

[5] Mikkel Thorup. “Undirected Single-Source Shortest Paths with Positive Integer Weights in Linear Time”. In: *Journal of the ACM* 46.3 (1999), pp. 362–394. DOI: 10.1145/316542.316548.

[6] Ran Duan et al. “Breaking the Sorting Barrier for Directed Single-Source Shortest Paths”. In: (2025), pp. 36–44. DOI: 10.1145/3717823.3718179.

[7] Till Tantau. *The Beamer Class User Guide*. 2004.

[8] University of the Philippines. *UP Visual Identity Guide 2017*. Accessed on April 2, 2026. URL: https://up.edu.ph/wp-content/uploads/2017/03/VIG-layout-Jan-2017B-compressed.pdf.